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MISS - report 1

Body moving in a uniform gravitational field

1. Approach

In our approach, we used formulas for $x(t)$ and $y(t)$ to calculate time, at which our points would cross the x/y coordinate of the Region line. Depending on whether it's vertical or horizontal, the x or y value is calculated and checked if it is contained in that region.

2. Charts

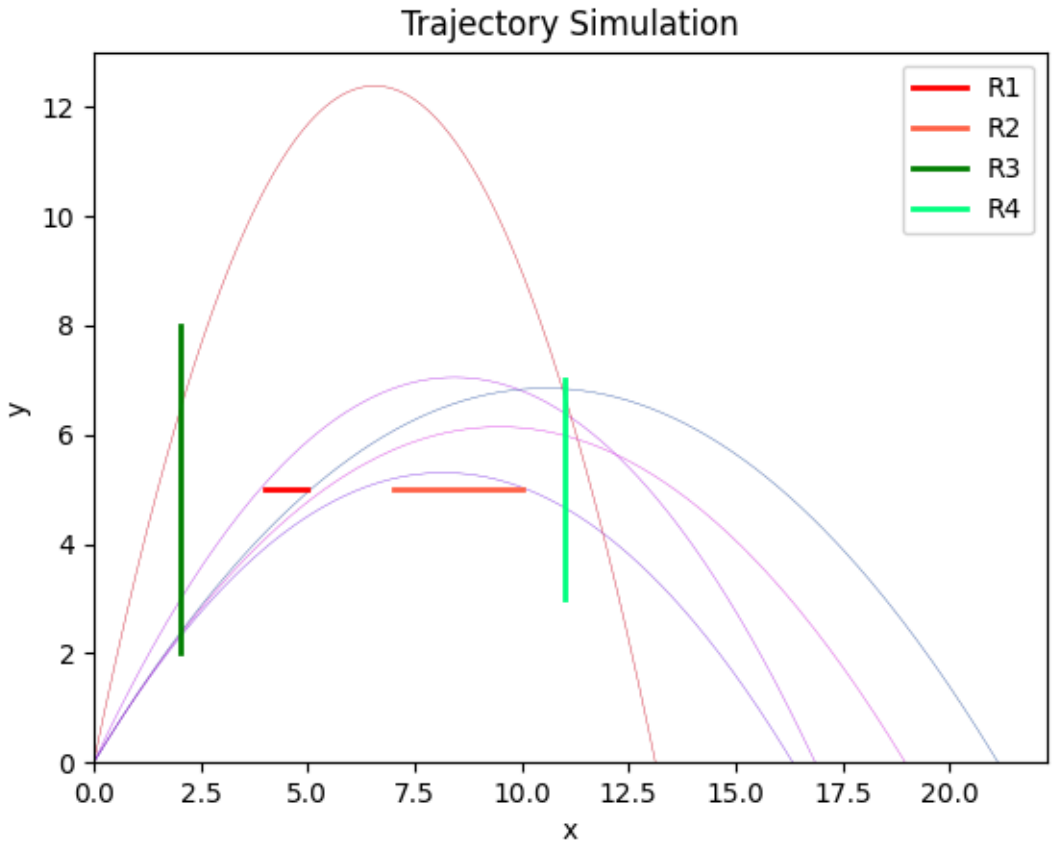


Figure 1: Example valid paths of the moving object passing through allowed (green) and disallowed (red) regions.

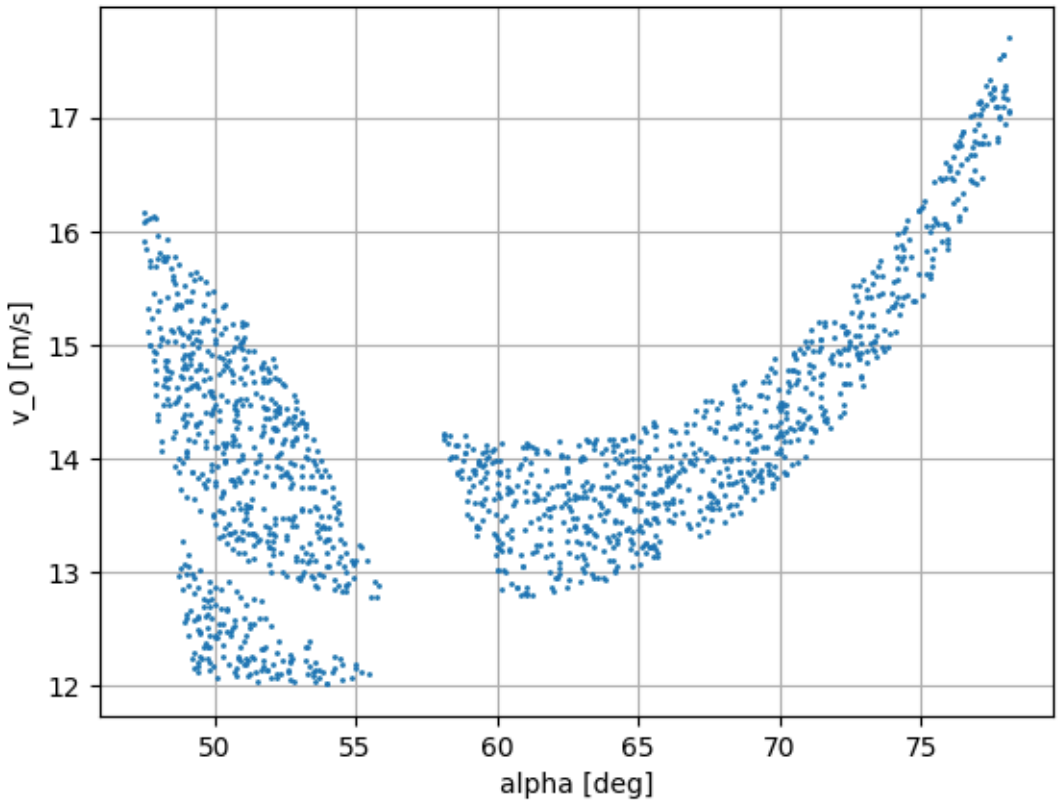


Figure 2: Plot of all values α and v_0 that satisfy the requirements about region crossing.

3. Probability of fulfilling the requirements

Given our regions and boundaries of α and v_0, our probability that the randomly selected values of α and v_0 met the requirements (about crossing the regions) was about **1.5671%** (for 1 million simulations).

The most important factor that affects the accuracy of the probability is the number of conducted simulations. The more values of α and v_0 are covered, the more our result approaches the true value of the probability.

Another aspect is the distribution of the random values generated which should be uniform. Otherwise, only some values of α and v_0 will be checked.

The size and location of the regions also affects the probability as there could be some extreme regions given as initial conditions for which the probability would converge more slowly with number of simulations.